Hypothesis Testing

Null and alternative hypotheses, Type I and II errors, significance and power, critical regions, tails and p-values.

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1 Hypotheses

A **statistical hypothesis** is a testable statement about a population parameter or distribution.

H_0: null

No effect, no difference, no relationship. The one tested for possible rejection.

H_1 / H_a: alternative

An effect, difference or relationship. Decides one- or two-tailed.

Karl Popper's **falsification principle**: a scientific hypothesis must be testable and potentially falsifiable.

2 Errors in decisions

Decision	H_0 true	H_0 false
Reject H_0	Type I error (α)	Correct: power $1-\beta$
Fail to reject	Correct ($1-\alpha$)	Type II error (β)

Level of significance

$$P(\text{reject } H_0 \mid H_0 \text{ true}) \le \alpha$$

Power

$$1-\beta$$

Larger samples generally increase power. α is fixed **before** the test (usually 0.05 or 0.01).

3 Regions & tails

Test statistic (Z, t, χ^2, F) → compared with its **sampling distribution** under H_0. The **critical value** separates the **rejection (critical) region** from the **acceptance region**.

H_1	Test	Reject H_0 when	α in tail
$\mu \neq \mu_0$	**Two-tailed**	Statistic below the lower or above the upper critical value	$\alpha/2$ each
$\mu > \mu_0$	**Right-tailed**	Statistic > critical value	α
$\mu < \mu_0$	**Left-tailed**	Statistic < critical value	α

4 The p-value

Probability, assuming H_0 true, of a statistic **as extreme or more extreme** than observed.

p ≤ α	Reject H_0
p > α	Fail to reject H_0

Small p = strong evidence against H_0.

5 The eight steps of a test

1. State H_0
2. State H_1
3. Set the significance level α
4. Collect the data
5. Calculate the test statistic
6. Find the critical value or region
7. Compare the statistic with it
8. Make and state the decision

6 History

Fisher's tea-tasting experiment: a lady claimed she could tell whether milk or tea was poured first. It's the classic illustration of significance testing.

! Common mistakes

× Saying "accept H_0": say **fail to reject** · Failing to reject does not prove H_0 true · Choosing α after seeing the data · Picking the tail from the result instead of from H_1.

? Fill in the blanks

1. Declaring a new fertiliser effective when it actually has no effect is a Type ____ error. [I]
2. If β = 0.2, the power of the test is ____. [0.8]
3. Testing whether a variety yields more than 40 q/ha needs a ____-tailed test. [right]
4. With p-value 0.03 and α = 0.05, we ____ H_0. [reject]
5. With p-value 0.08 and α = 0.05, we ____ H_0. [fail to reject]
6. In a two-tailed test at α = 0.05, each tail has area ____. [0.025]

Source: *Textbook of Agricultural Statistics*, Ch. 13 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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